SYSTEM PROPOSAL

SkyPlan: Weather-Based Activity Planner

Platform: Web Application

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1. Introduction

Weather conditions heavily influence people's daily decisions, especially when planning outdoor activities such as jogging, hiking, picnics, or sports. However, most people rely on separate weather apps and personal guesswork to decide whether an activity is feasible, often resulting in cancelled plans or discomfort.

Currently, users check weather forecasts on one platform and manually decide what activities are appropriate. This process is fragmented, time-consuming, and does not provide personalized recommendations based on the user's interests or location.

The proposed system, SkyPlan, is a web-based application that combines real-time weather data with an activity recommendation engine. It will allow users to input a city and date, retrieve accurate weather forecasts via a third-party API, and automatically suggest suitable indoor or outdoor activities based on both weather conditions and the user's chosen interests. The system will also allow users to save favorite locations and view their search history.

The system will integrate the OpenWeatherMap API to provide real-time and multi-day forecast data, demonstrating meaningful API integration alongside a rule-based recommendation engine.

2. Purpose of the Project

This proposal aims to present a web application that will address the fragmented process of checking weather and deciding on activities, streamline activity planning based on real-time forecasts, and improve user convenience and decision-making for general users, travelers, and outdoor enthusiasts.

3. Objectives

3.1 General Objective

To design and develop a web-based weather and activity planning system that provides users with accurate forecasts and personalized activity recommendations.

3.2 Specific Objectives

- To develop a secure, role-based user registration and authentication system.
- To design a system that retrieves real-time weather data from the OpenWeatherMap API.
- To provide multi-day forecast viewing for better activity planning ahead of time.
- To automate activity recommendations based on weather conditions and user-selected interest tags.
- To provide registered users with the ability to save favorite cities and view search history.
- To allow admins to configure activity-recommendation rules.
- To generate a dashboard of usage statistics and popular searches for admin review.
- To develop a responsive and user-friendly web interface.

4. Project Scope

4.1 In Scope

- User registration and login with role-based access (user, admin, guest)
- Profile and interest-tag preferences
- City and date search with weather forecast display
- Multi-day forecast view for trip/activity planning
- Activity recommendation engine (rule-based on weather conditions and interest tags)
- Save favorite cities and view search history

- Guest mode with limited (non-saving) search
- Admin dashboard for usage statistics and search logs
- Admin configuration of activity-recommendation rules
- Responsive web interface

4.2 Out of Scope

- Native mobile application (iOS/Android)
- Hardware procurement or IoT weather sensors
- Paid third-party API subscriptions (free tiers only)
- Content migration from other weather platforms
- SMS or push notification services
- Real-time GPS tracking
- Activity booking or reservation integration

5. System Features / Functions

Module	Feature	Description
User Management	Registration & Login	Allows users to create accounts and securely log in (with role-based access).
User Management	Profile & Interests	Users can view and update personal information and interest tags.
Core Module	Weather Search	Users enter a city and date to retrieve forecast data from the weather API.
Core Module	Multi-Day Forecast	Displays an extended forecast for planning activities ahead of time.
Core Module	Activity Recommendation	System suggests indoor/outdoor activities based on weather conditions and interest tags.
Core Module	Favorites & History	Users can save favorite cities and view past searches.
Guest Access	Limited Search	Allows unregistered visitors to check weather without saving favorites or history.
Reporting	Dashboard & Reports	Admin views user activity, popular cities, and search statistics.
Administration	Activity Rule Configuration	Admin configures the rules used by the recommendation engine.
Administration	System Settings	Admin manages user accounts and system parameters.

6. Technology Stack

Layer	Technology / Tool	Purpose
Frontend (Web)	React.js + Tailwind CSS	User interface for the web application

Layer	Technology / Tool	Purpose
Backend / API	Node.js + Express	Business logic and server-side processing
Database	PostgreSQL	Data storage and management
External API	OpenWeatherMap API	Real-time and multi-day weather data
Authentication	JWT	Secure authentication and role-based authorization
Charts	Chart.js	Visualizes weather trends and admin usage statistics
Cloud / Hosting	Vercel (frontend) + Render (backend)	Deployment and hosting environment
Version Control	Git / GitHub	Source code management
Project Management Tool	Trello	Task tracking and team collaboration
Other Tools	Figma, Postman	UI/UX design and API testing

7. Target Users

User Role	Description	Key Needs / Usage
Registered User	General users, travelers, outdoor enthusiasts	Search weather, receive activity suggestions, save favorites
Guest	Unregistered visitor	Limited search (no saving or history)
Admin	System administrator managing overall operations	Full access to manage users, activity rules, and reports

8. Project Timeline / Work Plan

Phase	Activities	Estimated Duration
Planning & Requirements Gathering	Finalize proposal, gather requirements, research OpenWeatherMap API.	1–2 weeks
System Design	Database schema, UI/UX wireframes, system architecture.	1–2 weeks
Development – Phase 1	Authentication, weather search, profile management.	2 weeks
Development – Phase 2	Multi-day forecast, favorites, and search history.	2 weeks
API Integration	OpenWeatherMap API integration and activity-recommendation engine.	1 week
Midterm Progress Check	Demonstration of approximately 60–70% functional system.	Milestone

Phase	Activities	Estimated Duration
Advanced Development	Admin rule configuration, dashboard, interface refinement.	1–2 weeks
Testing & QA	Unit, integration, and user acceptance testing; bug fixing.	1–2 weeks
Deployment	Cloud deployment and production testing.	1 week
Documentation	System documentation, screenshots, presentation materials.	1 week
Final Presentation	System demonstration and project defense.	Final Phase